

# LÊ NAM KHÁNH

AI / ML ENGINEERING | MULTIMODAL RETRIEVAL | RESEARCH

Information Technology student | University of Science, VNU-HCM (HCMUS)

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|-------------------------------|-----------------|-------------------|--------------------|
| AI CHALLENGE 2026             | HCMUS CODING    | CODEFORCES        | SOICT 2026         |
| Finalist   Bảng A (Top Teams) | Champion   2026 | Expert   Max 1796 | Paper First Author |

## PROFILE

Information Technology student at HCMUS with a rigorous competitive programming background and practical expertise in multimodal video retrieval and AI systems. First & corresponding author of SOICT 2026 research on structured temporal event alignment (GEMTRA) and Finalist at the Ho Chi Minh City AI Challenge 2026. Skilled in architecting end-to-end retrieval pipelines, fast algorithmic reasoning under constraints, and full-stack engineering.

## EDUCATION

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| <b>University of Science, VNU-HCM (HCMUS)</b><br>Information Technology   Second-year student | <b>Current direction</b><br>Multimodal video retrieval, temporal event reasoning, and foundation model systems. |
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## TECHNICAL SKILLS

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|-----------------------|---------------------------------------------------------------------------------|
| <b>Algorithms</b>     | Data structures, graph algorithms, dynamic programming, constraint optimization |
| <b>Languages</b>      | Python, C++, TypeScript / JavaScript, Rust, SQL, Bash / PowerShell              |
| <b>AI / Retrieval</b> | Visual embeddings (SigLIP2), FAISS, RF-DETR, OCR/ASR, VQA, GEMTRA alignment     |
| <b>Engineering</b>    | React 19 / Vite, Firebase / Firestore, Tauri, FastAPI, Git, testing, DRES API   |

## ACHIEVEMENTS

|                                                                                |                                                                         |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| AI Challenge HCMC 2026<br>Finalist (Bảng A)   Team Lead, Reply 404             | HCMUS Coding Challenge<br>Champion   2026                               |
| SOICT 2026 Full Paper<br>First & Corresponding Author (LNCS)                   | Codeforces<br>Expert   Max rating 1796                                  |
| National Young Informatics<br>First Prize (Central)   Nat'l Honourable Mention | Provincial & Olympic 30/4<br>Provincial 2nd & 3rd Prizes   Bronze Medal |

## RELEVANT STRENGTHS

|                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                               |
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| <ul style="list-style-type: none"><li>- Fast at translating mathematical formulations &amp; constraints into efficient implementations.</li><li>- Resilient operator with high debugging agility under live competition countdowns.</li><li>- Rigorous engineering mindset: separating empirical evidence from benchmark claims.</li></ul> | <ul style="list-style-type: none"><li>- Full-stack capability: bridging ML/AI backend systems with fast, intuitive user experiences.</li><li>- Strong team collaborator across research ideation, system benchmarking, and code review.</li><li>- English: IELTS 6.5   GitHub: lenamkhanhh   Codeforces: Average2k7</li></ul> |
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## SELECTED PROJECTS

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| <b>Reply 404 — Video Retrieval System</b><br>Team Lead & Core Architect   Python, SigLIP2, GEMTRA DP, DRES API<br>Led 4-member team to design and build an interactive video search platform for 1,487 videos & 533K keyframes with sub-7ms temporal DP alignment (GEMTRA) and evidence-linked VQA under AIC countdowns.<br>SOICT 2026 Paper   GitHub / HCMAIC-Retrieval |
| <b>TripFlow Workbench</b><br>React 19   TypeScript   Vite   Firebase Auth & Firestore Rules<br>Built a collaborative travel planning workbench featuring 4 realtime workflows: timeline scheduling with priority/assignees, expense splitting, and team member synchronization.<br>Live App / mxhuit26.vercel.app   GitHub                                               |
| <b>AI Account Switcher</b><br>Tauri   Rust   React   Local API Gateway   CLI Wrappers<br>Desktop app managing multi-account rotation for AI coding assistants (Claude Code, Codex) with an OpenAI-compatible local proxy gateway and real-time quota tracking.<br>GitHub / ai-switcher                                                                                   |
| <b>RESEARCH-ORIENTED INTERESTS</b><br>Multimodal Video Retrieval   Temporal Event Reasoning   Vision-Language Models   AI Agents<br>Also building competitive programming training artifacts and reproducible benchmarks.                                                                                                                                                |